9/1/2026

N + - Network Fundamentals & Building Networks.

IP addressing & Subnetting.

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TASK 1.

❖ **List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.**

➢ **IP Address Classes:**

IP Address Class	Address Range	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	Large organizations such as ISPs
Class B	128.0.0.0 – 191.255.255.255	University networks
Class C	192.0.0.0 – 223.255.255.255	Small office or home networks
Class D	224.0.0.0 – 239.255.255.255	Multicast services such as video streaming
Class E	240.0.0.0 – 255.255.255.255	Experimental and research networks

TASK 2.

❖ Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses, Show your working steps for each calculation.

➢ **IP Address and Subnet Calculation:**

Given:

- IP Address**: 192.168.10.45**
- Subnet Mask**: 255.255.255.0**

ℵ **Network Address**

Subnet mask **255.255.255.0** means the first three octets represent the network.

- **192.168.10.45**
  AND **255.255.255.0**
  = 192.168.10.0

- **Network Address = 192.168.10.0**

ℵ **Broadcast Address**

Since the last octet is the host portion, all host bits are set to **1**:

- **192.168.10.255**
- **Broadcast Address = 192.168.10.255**

ℵ **Valid Host Address Range**

The first address **(192.168.10.0)** is reserved for the network, and the last **(192.168.10.255)** is reserved for broadcast.

Therefore:

- **Valid Host Range = 192.168.10.1 – 192.168.10.254**

TASK 3.

❖ **Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.**

➢ **Subnetting Plan for a Food Delivery Startup:**
Using the **10.0.0.0/24** network, it can be divided into **four equal /26 subnets**.
Each subnet provides **62 valid host addresses**.

Group	Network Address	Subnet Mask
Delivery Partners	10.0.0.0/26	255.255.255.192
Restaurants	10.0.0.64/26	255.255.255.192
Customers	10.0.0.128/26	255.255.255.192
Admins	10.0.0.192/26	255.255.255.192

TASK 4.

- **You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet, Show your calculation steps and final answers.**

- **Subnetting a College WiFi Network:**

Given:

- IP Network**: 172.16.0.0/16**
- Required Subnets**: At least 20**

ℵ **Calculate the New Subnet Mask**

We need to borrow bits from the host portion:

- **2^4 = 16 →** not enough
- **2^5 = 32 →** enough

So, **5 bits** are borrowed.

- **New Prefix = /16 + 5 = /21**
- **Subnet Mask = 255.255.248.0**

ℵ **Calculate Hosts per Subnet**

Remaining host bits:

- **32 − 21 = 11 bits**

Total addresses per subnet:

- **2^{11} = 2048**

Usable host addresses:

- **2048 − 2 = 2046**

Final Answers

- **New Subnet Mask: 255.255.248.0 (/21)**
- **Number of Subnets: 32**
- **Usable Hosts per Subnet: 2046**

TASK 5.

- **A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.**

- **Invalid IP Address:**
  The IP address **192.168.1.300** is invalid because each part of an IPv4 address must be between **0 and 255**. Here, **300** exceeds the maximum allowed value.

  A valid IP address in the **192.168.1.0/24** subnet should have the form **192.168.1.x**, where **x ranges from 1 to 254**.

  Example of a valid IP address: 192.168.1.100